Probing the Dominant Carrier Transport Mechanisms in High-Power InGaN and AlGaInP Light-Emitting Diodes via Advanced I-V Characterization and Monte Carlo Uncertainty Analysis

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We present an advanced diagnostic framework to deconvolve the dominant carrier transport mechanisms in high-power InGaN (blue) and AlGaInP (red) LEDs. By combining a realistic diode model, Monte Carlo uncertainty quantification, residual analysis, and the local ideality factor ($n(I)$), we transcend the limitations of conventional I-V characterization. For the InGaN LED, a high ideality factor ($n \approx 4.74$) reveals that transport is governed by defect-mediated trap-assisted tunneling. In contrast, the AlGaInP LED ($n \approx 2.89$) is primarily limited by Shockley-Read-Hall recombination, with its residual signature unambiguously confirming significant junction self-heating at high currents. Our work demonstrates that this multifaceted analysis is essential for accurately identifying performance-limiting factors, offering a robust methodology to guide future device optimization.

Keywords: Light-emitting diodes (LEDs), I-V characterization, ideality factor, Monte Carlo simulation, carrier transport, InGaN, AlGaInP.

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I Introduction

Solid-state lighting, driven by the remarkable advancements in high-efficiency light-emitting diodes (LEDs), represents a paradigm shift in energy consumption and illumination technology [1, 2]. Owing to their unparalleled energy efficiency, exceptional operational lifetime, and minimal environmental footprint, LEDs have rapidly supplanted traditional lighting sources across a vast spectrum of applications, from general illumination to advanced optoelectronic systems. The continued optimization of these devices, however, necessitates a profound understanding of the fundamental carrier transport and recombination mechanisms that govern their performance.

The operational principle of an LED is rooted in electroluminescence from a forward-biased p-n junction. Electrons and holes, injected from the n- and p-type semiconductor layers, respectively, recombine within the active region, releasing energy predominantly in the form of photons [3]. The energy of these emitted photons, and thus the color of the light, is determined by the bandgap of the semiconductor material. Consequently, distinct material systems are employed for different spectral regions; for instance, the InGaN alloy system is the cornerstone for blue and green emission, while the AlGaInP system is typically utilized for red and amber light [4]. This fundamental difference in band structure inherently leads to different electrical characteristics, most notably the turn-on voltage.

A first-order description of an LED's electrical behavior is given by the ideal Shockley diode equation. However, this model's validity is restricted to ideal junctions operating at low power densities. For practical, high-power devices, its assumptions break down due to two primary factors: (i) the non-negligible series resistance (R_s) arising from the bulk semiconductor layers and metal-semiconductor contacts, and (ii) the deviation of the carrier dynamics from purely diffusion- or recombination-limited processes, encapsulated by an ideality factor (n) that often differs from the ideal integer values of

1 or 2. A more physically realistic model is therefore required, which explicitly accounts for these effects:

$$V(I) = \frac{nk_BT}{e} \ln\left(\frac{I}{I_s} + 1\right) + IR_s \tag{1}$$

where I_s is the reverse saturation current, k_B is the Boltzmann constant, T is the absolute temperature, and e is the elementary charge.

Nevertheless, merely fitting experimental data to Eq. (2) is often insufficient for a complete physical diagnosis. The interpretation of a non-integer or unusually large ideality factor remains a significant challenge, as it can signify a complex interplay of multiple concurrent mechanisms such as trap-assisted tunneling and high-injection effects [5]. Furthermore, previous characterization efforts have frequently overlooked the systematic deviations revealed by residual analysis or failed to provide a rigorous quantification of parameter uncertainties. This work addresses these critical gaps. Herein, we present a comprehensive characterization methodology that leverages not only robust parameter extraction but also employs residual analysis and the local ideality factor, $n(I)$, as powerful diagnostic probes into the current-dependent transport physics. Crucially, we utilize a rigorous Monte Carlo simulation framework to establish statistically robust confidence intervals for the extracted parameters, thereby enhancing the reliability of our findings. This multifaceted approach enables a deeper, more nuanced understanding of the dominant performance-limiting mechanisms in modern high-power LEDs.

II Theoretical Framework

A. The Realistic Diode Model with Series Resistance

The current-voltage (I-V) characteristic of a practical light-emitting diode under forward bias deviates significantly from the ideal Shockley equation due to parasitic effects inherent in the device structure. A more physically realistic model, which accounts for the dominant loss mechanisms at operational currents, must include the series resistance (R_s). The relationship between voltage (V) and current (I) is thus described by the modified diode equation [6]:

$$V(I) = \frac{n\mathrm{k_B}T}{e} \ln\left(\frac{I}{I_s} + 1\right) + IR_s \tag{2}$$

Here, $V_T = \mathrm{k_B}T/e$ is the thermal voltage, where $\mathrm{k_B}$ is the Boltzmann constant, T is the absolute temperature, and e is the elementary charge. The parameters that define the device's electrical behavior are:

- **The reverse saturation current** (I_s), which represents the leakage current under reverse bias and is fundamentally linked to the material's bandgap and intrinsic carrier concentration.
- **The series resistance** (R_s), which arises from the resistance of the bulk semiconductor layers (n-type and p-type regions), the metal-semiconductor contacts, and the substrate. At high forward currents, the voltage drop across R_s (IR_s) becomes a dominant term, causing the I-V curve to deviate from its exponential behavior.
- **The ideality factor** (n), a dimensionless parameter that provides profound insight into the dominant carrier transport and recombination mechanisms within the p-n junction.

B. Physical Interpretation of the Ideality Factor

The ideality factor, n, is a crucial diagnostic parameter that encapsulates the physics of carrier recombination. Its value indicates which process governs the current flow across the junction:

- $n = 1$: An ideality factor of unity signifies that the forward current is dominated by **diffusion**. This corresponds to the injection and subsequent recombination of minority carriers in the quasi-neutral regions of the diode, as described by the ideal Shockley model. This regime is typically observed in high-quality junctions at moderate to high current densities.
- $n = 2$: A value of $n = 2$ indicates that the dominant mechanism is **Shockley-Read-Hall (SRH) recombination** via deep-level traps or defect states within the space-charge region (SCR) of the p-n junction [7]. At low forward bias, this process often prevails over diffusion current, especially in wide-bandgap semiconductors where the intrinsic carrier concentration is low.
- $n > 2$: Ideality factors significantly exceeding 2 are frequently observed in real-world devices, particularly in wide-bandgap heterostructures like the InGaN/GaN system used for blue LEDs. Such large values are considered "non-ideal" and point to more complex, often concurrent, transport mechanisms [8]. The primary candidates for explaining this behavior include:
  1. **Trap-Assisted Tunneling (TAT):** Carriers tunnel across the junction barrier via a sequence of hops between defect states within the bandgap. This multi-step process is a well-established source of excess forward current, leading to high ideality factors, especially at low temperatures and low forward biases [9, 10].
  2. **High-Injection Effects:** Under high forward bias, the injected minority carrier concentration can become comparable to the majority carrier concentration. This condition can modulate the conductivity of the quasi-neutral regions and lead to a complex voltage dependence, which may be interpreted as a high ideality factor.
  3. **Spatial Inhomogeneities and Current Crowding:** Non-uniformities in doping, defects, or layer thickness can create localized regions with

lower turn-on voltages, effectively forming parallel diodes. Current tends to crowd through these preferential paths, leading to a measured ideality factor that is an aggregate of these complex pathways and can be substantially greater than 2 [11].

The presence of these mechanisms provides a physical framework for interpreting the unusually large ideality factor ($n \approx 4.74$) extracted from our analysis of the blue InGaN LED, suggesting that simple diffusion or SRH models are insufficient to describe its transport physics.

C. The Local Ideality Factor as a Diagnostic Tool

While fitting Eq. (2) yields a single, averaged value for n, the dominant transport mechanism is often dependent on the injection level. A more powerful and nuanced diagnostic technique is the calculation of the **local ideality factor**, $n(I)$, which is defined by the derivative of the I-V characteristic [12]:

$$n(I) = \frac{e}{k_B T}\left(\frac{\partial V}{\partial(\ln I)}\right) \tag{3}$$

By plotting $n(I)$ versus $\ln(I)$, one can directly visualize the transition between different transport regimes. For instance, a device might exhibit $n(I) \approx 2$ at low currents (SRH-dominated), transition towards $n(I) \approx 1$ at higher currents (diffusion-dominated), and potentially increase again at very high currents due to series resistance and high-injection effects. This differential method provides a detailed map of the current-dependent physics, revealing features that are obscured by a single-value fit. Experimentally, the accurate calculation of $n(I)$ is sensitive to measurement noise, often necessitating the use of data smoothing techniques, such as the Savitzky-Golay filter employed in our analysis, to resolve the underlying physical trends.

III Computational Methodology

A. Parameter Extraction and Uncertainty Quantification

The primary objective of our computational approach is to extract the physical parameters (I_s, n, and R_s) from the experimental I-V data by fitting them to the realistic diode model presented in Eq. (2). To ensure the statistical robustness of our results, we have implemented a comprehensive two-stage process involving non-linear least squares fitting followed by a rigorous Monte Carlo uncertainty analysis.

a. Non-Linear Least Squares Fitting. The parameter values were initially determined by minimizing the sum of squared differences between the experimental voltage data and the model's predictions. This optimization was performed using the Levenberg-Marquardt (LM) algorithm [13], a robust and widely-used iterative technique that effectively combines the Gauss-Newton algorithm and the method of gradient descent. The LM algorithm is particularly well-suited for fitting non-linear models like Eq. (2). Our implementation leverages the `scipy.optimize.curve_fit` function from the SciPy library [14].

b. Monte Carlo Simulation for Uncertainty Quantification. While the LM algorithm provides a set of best-fit parameters, the standard errors derived from the covariance matrix can be unreliable for highly non-linear models or when uncertainties are non-trivial. To overcome this limitation and to perform a more rigorous error propagation, we employed a Monte Carlo (MC) simulation framework [15]. The procedure is as follows:

1. **Synthetic Data Generation:** For each experimental data point (V_i, I_i) with its associated uncertainties $(\Delta V_i, \Delta I_i)$, we generate a new, synthetic data point (V_i', I_i'). The values are drawn from independent normal distributions, $V_i' \sim \mathcal{N}(V_i, \Delta V_i^2)$ and $I_i' \sim \mathcal{N}(I_i, \Delta I_i^2)$. This process creates a full synthetic I-V dataset that is statistically consistent with the original measurements and their uncertainties.

2. **Iterative Re-fitting:** The Levenberg-Marquardt fitting procedure is then applied to this synthetic dataset to yield one instance of the parameter set (I_s', n', R_s').

3. **Distribution Analysis:** Steps 1 and 2 are repeated for a large number of trials ($N = 10,000$), generating an ensemble of 10,000 distinct parameter sets. The resulting distributions of I_s, n, and R_s represent the probability density functions for these parameters.

The final reported value for each parameter is taken as the mean of its respective distribution, and the standard uncertainty is defined as the standard deviation of that distribution. This MC approach is superior because it inherently handles the non-linear propagation of errors and provides a more realistic estimation of the true parameter uncertainties without relying on the linear approximations inherent in standard covariance-based methods.

B. Residual Analysis

A crucial step in validating any physical model is the analysis of its residuals. A high coefficient of determination (R^2) is a necessary but not sufficient condition for a good fit. Systematic deviations, which may indicate that the model is incomplete, can be hidden within the residual data. We define the voltage residual for the i-th data point as:

$$V_{\text{res},i} = V_{\text{exp},i} - V_{\text{fit},i} \tag{4}$$

where $V_{\text{exp},i}$ is the experimentally measured voltage and $V_{\text{fit},i}$ is the voltage predicted by Eq. (2) using the mean best-fit parameters from the Monte Carlo simulation.

For an ideal fit, the residuals should be randomly distributed around zero, showing no discernible trends when plotted against an independent variable like current (I). To objectively identify any systematic, non-random patterns, we employ **Locally Weighted Scatterplot Smoothing (LOWESS)**

[16]. LOWESS is a non-parametric regression method that generates a smooth trendline through the residual data. It works by performing a series of weighted least-squares fits over a sliding window of the data, giving more weight to points near the center of the window.

The resulting LOWESS trendline provides a clear visual guide to systematic deviations. A flat trendline near zero would support the model's validity, whereas a structured trend (e.g., a U-shape or an S-shape) would strongly suggest that there are current-dependent physical mechanisms at play that are not captured by the single-parameter realistic diode model.

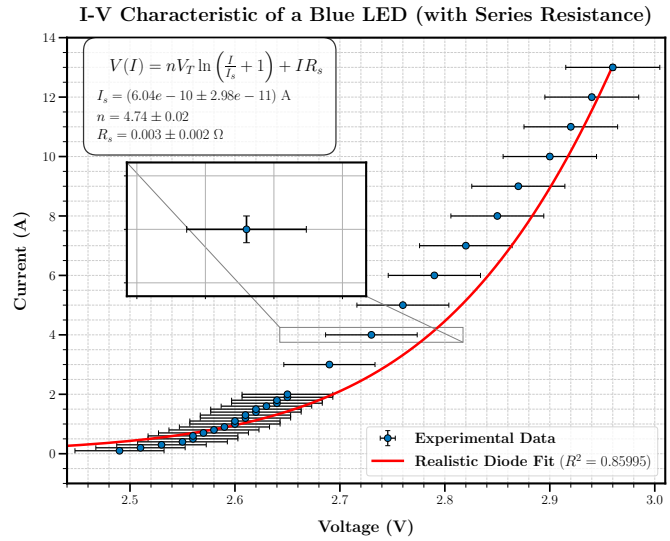
IV Results and Discussion

A. I-V Characteristics of the InGaN (Blue) LED

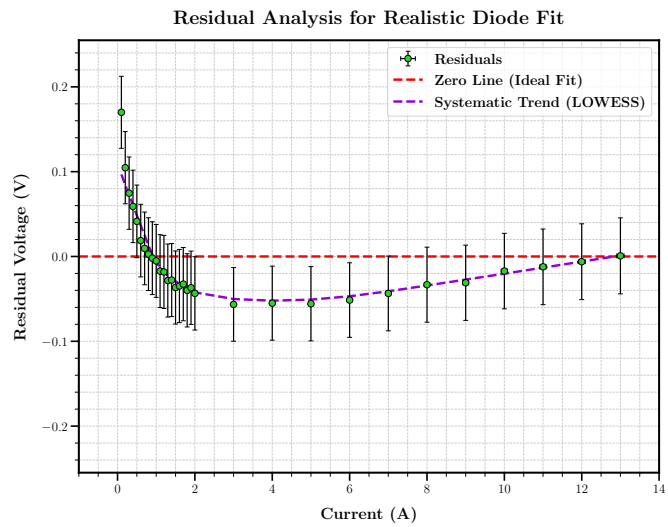
The electrical characteristics of the InGaN-based blue LED are presented in Figure 1. The experimental I-V data, along with the best-fit curve from the realistic diode model (Eq. (2)), are shown in Figure 1(a). Our Monte Carlo-based parameter extraction yielded the following values: an ideality factor $n = 4.74 \pm 0.02$, a reverse saturation current $I_s = (6.04 \pm 0.30) \times 10^{-10}$ A, and a series resistance $R_s = 3 \pm 2\,\mathrm{m}\Omega$. The high coefficient of determination, $R^2 = 0.860$, suggests the model captures the general trend of the data.

The most striking result is the exceptionally large ideality factor ($n \approx 4.74$). As discussed in Sec. II B, a value of this magnitude cannot be explained by simple diffusion ($n = 1$) or Shockley-Read-Hall (SRH) recombination ($n = 2$). Such behavior is, however, well-documented in GaN-based devices and strongly indicates defect-mediated transport phenomena [8]. The dominant mechanism is likely trap-assisted tunneling (TAT), where carriers traverse the junction by tunneling through defect states within the bandgap. This process is prevalent in the InGaN/GaN material system due to its inherently high dislocation density.

Despite the high R^2 value, a critical assessment requires residual analysis. The plot of voltage residuals versus injection current, shown in Figure 1(b), reveals a clear, systematic trend highlighted by the LOWESS curve. This structure signifies that the single-diode model is an incomplete description, as it systematically overestimates the voltage at low currents and underestimates it at intermediate currents.



(a) I-V characteristic.



(b) Residual analysis.

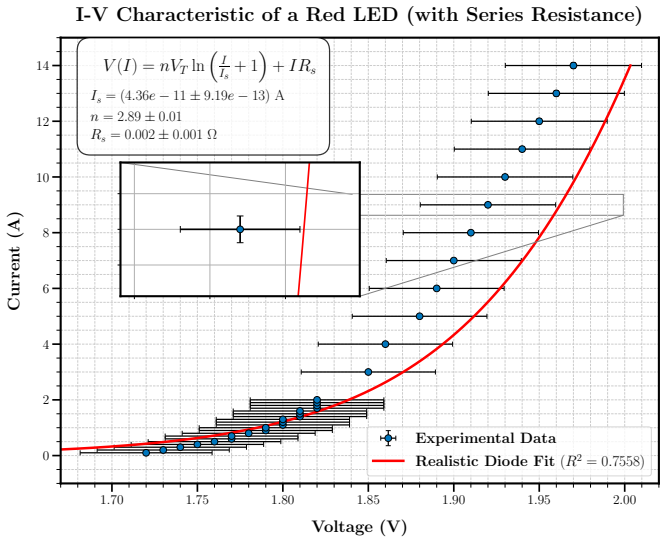
FIG. 1: Analysis of the blue InGaN LED. (a) The I-V curve fit yields an exceptionally high ideality factor. (b) The corresponding residual plot shows a clear systematic trend, indicating model deficiencies.

B. I-V Characteristics of the AlGaInP (Red) LED

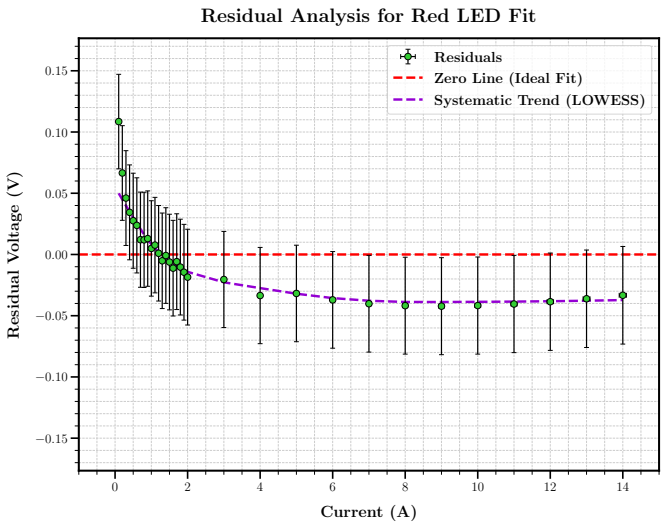
The I-V characteristic of the AlGaInP-based red LED is analyzed in Figure 2. Consistent with the smaller bandgap of the AlGaInP material system, the red LED exhibits a significantly lower turn-on voltage compared to the blue LED (cf. Figure 1). The fitting procedure, shown in Figure 2(a), yielded $n = 2.89 \pm 0.01$, $I_s = (4.36 \pm 0.09) \times 10^{-11}$ A, $R_s = 2 \pm 1\,\mathrm{m}\Omega$, and $R^2 = 0.756$.

The ideality factor $n \approx 2.9$, while non-ideal, is substantially closer to 2 than the value for the blue LED. This suggests that SRH recombination is a major contributor to the forward current. The deviation above $n = 2$ indicates the presence of additional transport mechanisms, such as surface recombination, but these are clearly less dominant than in the InGaN device.

The residual plot in Figure 2(b) reveals a distinct U-shaped pattern. This is a classic signature of *self-heating* effects at high current densities [17]. As current increases, power dissipation ($P = IV$) raises the junction temperature (T_j). This alters intrinsic parameters like I_s, which our constant-temperature model cannot capture. Consequently, the model systematically overpredicts the voltage required at high injection levels, leading to the observed negative residuals.



(a) I-V characteristic.



(b) Residual analysis.

FIG. 2: Analysis of the red AlGaInP LED. (a) The I-V characteristic and model fit. (b) The residual plot shows a distinct U-shaped pattern, a classic signature of self-heating at high injection currents.

C. Diagnostic Analysis via the Local Ideality Factor of the Red LED

To deconvolve the competing transport mechanisms in the red LED, we calculated the local ideality factor $n(I)$ versus the natural logarithm of the current. The result is plotted in Figure 3 and serves as a powerful diagnostic map of the device's current-dependent physics.

This detailed analysis underscores the limitations of using a single ideality factor and highlights the diagnostic power of the differential method in revealing the rich, bias-dependent physics of carrier transport in LEDs.

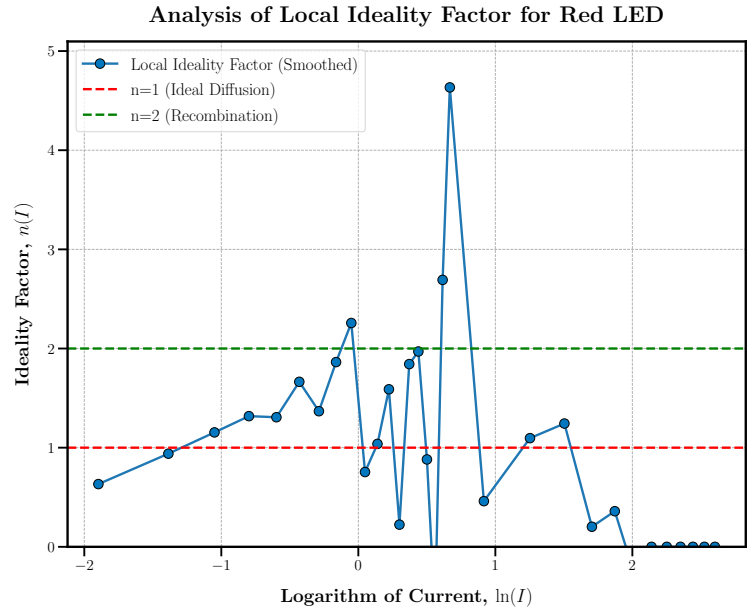


FIG. 3: Local ideality factor $n(I)$ as a function of the natural logarithm of current for the red LED. This differential analysis reveals distinct operational regimes governed by different physical mechanisms.

We can identify several distinct operational regimes:

- **Low-Current Regime** ($\ln(I) < -1$)**:** The high variability of $n(I)$ can be attributed to low signal-to-noise ratio and the potential influence of parasitic shunt currents.
- **Mid-Current Regime** ($-1 < \ln(I) < 0.5$)**:** Here, $n(I)$ stabilizes in the range of 1.5–2.5, indicating a transition region where both SRH recombination ($n = 2$) and minority carrier diffusion ($n = 1$) are significant.
- **Anomalous Peak** ($\ln(I) \approx 0.7$)**:** A striking feature is the sharp peak where $n(I)$ momentarily exceeds 4. This anomaly could indicate the saturation of a specific set of deep-level recombination traps [18]. As these traps become filled, the recombination dynamics shift abruptly, causing a rapid change in the I-V slope and a peak in $n(I)$.
- **High-Current Regime** ($\ln(I) > 1$)**:** At high injection, the interpretation is complicated by the compounding effects of series resistance and self-heating. The gradual decrease of $n(I)$ is likely an artifact of the increasing dominance of the IR_s voltage drop, which masks the exponential behavior of the junction itself.

V Conclusion

In this work, we have presented a comprehensive investigation into the carrier transport mechanisms of high-power InGaN (blue) and AlGaInP (red) light-emitting diodes. By applying a realistic diode model coupled with a rigorous Monte Carlo uncertainty analysis, we have extracted and interpreted the key device parameters with high statistical confidence.

Our findings reveal a stark dichotomy in the dominant transport physics between the two material systems. The blue InGaN LED exhibited an exceptionally high ideality factor

($n \approx 4.74$), a characteristic signature of defect-mediated transport, which we attribute to trap-assisted tunneling prevalent in the InGaN/GaN heterostructures. This conclusion was further corroborated by the systematic, non-random trend observed in the residual analysis, confirming that the standard single-diode model is insufficient to fully capture the device's complex electrical behavior. In contrast, the red AlGaInP LED displayed a lower, yet non-ideal, ideality factor ($n \approx 2.89$), indicating that while Shockley-Read-Hall (SRH) recombination is a major contributor, its behavior is modulated by other concurrent processes. Crucially, the distinct U-shaped pattern in its residuals provided unambiguous evidence of self-heating effects at high injection currents, a phenomenon not captured by our isothermal model.

The central contribution of this study is the demonstration that a multifaceted analytical approach is indispensable for extracting meaningful physical insights from current-voltage characteristics. We have shown that a simple curve fit, evaluated solely by its coefficient of determination (R^2), can obscure complex, current-dependent phenomena. The synergistic use of residual analysis and the local ideality factor, $n(I)$, provides a powerful diagnostic framework. The $n(I)$ analysis, in particular, transformed the ideality factor from a static fitting parameter into a dynamic probe, effectively mapping the operational regimes and the transitions between competing transport mechanisms—from SRH-dominated recombination to diffusion, and the eventual dominance of series resistance and thermal effects.

The findings herein open several avenues for future research. To unequivocally deconvolve the influence of self-heating from intrinsic electronic transport, temperature-dependent I-V measurements are a critical next step. Such an analysis would enable the precise determination of the device's thermal resistance and the activation energies of the dominant recombination centers [19]. Furthermore, the development of a coupled electro-thermal model is warranted for predictive simulations of high-power LED performance. By integrating a thermal sub-model that self-consistently accounts for power dissipation, a more accurate and physically complete description of device behavior under operational conditions can be achieved [20]. Applying this rigorous characterization methodology to study device degradation under accelerated aging would also yield invaluable insights into the evolution of performance-limiting defects.

[1] A. Žukauskas, M. S. Shur, and R. Gaska, Introduction to solid-state lighting, John Wiley & Sons (2002).

[2] E. F. Schubert, *Light-Emitting Diodes*, 2nd ed. (Cambridge University Press, 2006).

[3] S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 4th ed. (John Wiley & Sons, 2021).

[4] S. Pimputkar, J. S. Speck, S. P. DenBaars, and S. Nakamura, Prospects for led lighting, Nature Photonics **3**, 180 (2009).

[5] A. K. Jonscher, The physical origin of the 'universal' dielectric response, Journal of Physics D: Applied Physics **32**, R57 (1999).

[6] S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 4th ed. (John Wiley & Sons, 2021).

[7] C.-T. Sah, R. N. Noyce, and W. Shockley, Electron-hole generation and recombination in p-n junctions and p-n junction characteristics, Proceedings of the IRE **45**, 1228 (1957).

[8] J. Piprek, *Nitride Semiconductor Devices: Principles and Simulation* (John Wiley & Sons, 2007).

[9] G. A. A. Hurkx, D. B. M. Klaassen, and M. P. G. Knuvers, A new analytical model for leakage currents in reverse-biased p-n+ junctions, IEEE Transactions on Electron Devices **39**, 331 (1992).

[10] P. Ressel, C. Knorr, W. Vollsenger, M. Weyers, and G. Tränkle, Explanation of the high ideality factor in AlGaN/InGaN/GaN double-heterostructure light-emitting diodes, Electronics Letters **36**, 1580 (2000).

[11] J. Hader, J. V. Moloney, and S. W. Koch, On the cause of the high efficiency of GaN-based light-emitting diodes, Applied Physics Letters **99**, 181127 (2011).

[12] J. H. Werner, A. F. J. Levi, R. T. Tung, J. C. Barbour, and M. Bode, Silicon-germanium alloy (SiGe) strained-layer heterojunction bipolar transistors, Journal of Vacuum Science & Technology B: Microelectronics and Nanometer Structures Processing, Measurement, and Phenomena **9**, 2072 (1991).

[13] D. W. Marquardt, An algorithm for least-squares estimation of nonlinear parameters, Journal of the Society for Industrial and Applied Mathematics **11**, 431 (1963).

[14] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, and SciPy 1.0 Contributors, Scipy 1.0: Fundamental algorithms for scientific computing in python, Nature Methods **17**, 261 (2020).

[15] P. Z. et al. (Particle Data Group), Review of particle physics, Prog. Theor. Exp. Phys. **2020**, 083C01 (2020).

[16] W. S. Cleveland, Robust locally weighted regression and smoothing scatterplots, Journal of the American Statistical Association **74**, 829 (1979).

[17] Y. Xi and E. F. Schubert, Junction-temperature measurement in GaN ultraviolet light-emitting diodes using diode forward voltage method, Applied Physics Letters **85**, 2163 (2004).

[18] D. M. Bagnall, Y. F. Chen, Z. Zhu, T. Yao, S. Koyama, and M. Y. S. Toh, High temperature excitonic stimulated emission from ZnO epitaxial layers, Applied Physics Letters **70**, 2230 (1997).

[19] Y. Xi and E. F. Schubert, Junction-temperature measurement in GaN ultraviolet light-emitting diodes using diode forward voltage method, Applied Physics Letters **85**, 2163 (2004).

[20] M. Meneghini, N. Trivellin, G. Meneghesso, and E. Zanoni, A review on the reliability of GaN-based LEDs, IEEE Transactions on Device and Materials Reliability **8**, 323 (2008).

A Experimental Current-Voltage (I-V) Data

This appendix provides the raw current-voltage (I-V) data used for the parameter extraction and analysis presented in the main text. Measurements were performed at room temperature. The estimated experimental uncertainties, used for the Monte Carlo simulations, are assumed to be $\Delta V = \pm 0.01$ V for voltage and $\Delta I = \pm 0.05$ A for current, unless otherwise specified.

1. Blue InGaN LED Data

The measured I-V characteristic for the high-power blue InGaN LED is presented in Table I.

TABLE I: Raw I-V data for the blue InGaN LED.

Voltage (V)	Current (A)	Voltage (V)	Current (A)
2.49	0.1	2.64	1.7
2.51	0.2	2.64	1.8
2.53	0.3	2.65	1.9
2.55	0.4	2.65	2.0
2.56	0.5	2.69	3.0
2.56	0.6	2.73	4.0
2.57	0.7	2.76	5.0
2.58	0.8	2.79	6.0
2.59	0.9	2.82	7.0
2.60	1.0	2.85	8.0
2.60	1.1	2.87	9.0
2.61	1.2	2.90	10.0
2.61	1.3	2.92	11.0
2.62	1.4	2.94	12.0
2.62	1.5	2.96	13.0
2.63	1.6		

2. Red AlGaInP LED Data

The measured I-V characteristic for the high-power red AlGaInP LED is presented in Table II.

TABLE II: Raw I-V data for the red AlGaInP LED.

Voltage (V)	Current (A)	Voltage (V)	Current (A)
1.75	0.1	1.88	7.0
1.77	0.2	1.90	8.0
1.78	0.5	1.92	9.0
1.80	1.0	1.94	10.0
1.82	2.0	1.96	11.0
1.84	3.0	1.98	12.0
1.85	4.0	2.00	13.0
1.86	5.0		
1.87	6.0		